



Week 06 Task-17

Password Cracking - Hashes (John /Hashcat)

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Reference Number: TBS-INT-2026-445

Submission Date: 24/9/2026

Submitted To: TechBiz Security

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1) Executive Summary

This practical task focused on understanding password security mechanisms and evaluating the effectiveness of password hashing in protecting user credentials. The assessment was conducted in a controlled and authorized lab environment to examine how passwords are stored using cryptographic hash functions and how weak passwords can be recovered through password-cracking techniques. During the exercise, a sample password was hashed using the MD5 algorithm, and industry-standard tools, including John the Ripper and Hashcat in TryHackMe, were utilized to perform a dictionary-based password-cracking attack. The recovered password was successfully verified against the original hash, demonstrating how weak or commonly used passwords can be compromised despite being stored in hashed form. The task also explored the fundamental differences between hashing and encryption, emphasizing the role of secure password storage practices in modern information systems.

2) Scope of Task

The scope of this task was limited to evaluating password hashing and password security within a controlled and authorized lab environment. The assessment included Generating an MD5 hash of a sample password to understand how passwords are represented using cryptographic hash functions. Performing dictionary-based password-cracking exercises using John the Ripper and Hashcat within the TryHackMe lab environment. Verifying the recovered password against the original hash to demonstrate the risks associated with weak or commonly used passwords. Understanding the fundamental differences between hashing and encryption and their respective purposes in information security. The assessment was restricted to sample passwords and authorized lab resources. No real user credentials, unauthorized systems, or third-party password databases were targeted.

3) Tools Used

- John the Ripper
- rockyou.txt (wordlist)

4) Explaining Password Hashing

Password hashing is the process of converting a password into a fixed-length string of characters, known as a hash, using a mathematical hashing algorithm. The generated hash is unique for a specific input and cannot be easily converted back to the original password. When a user creates an account, the system stores the hashed version of the password instead of the actual password. During login, the entered password is hashed again and compared with the stored hash. If both hashes match, the user is authenticated successfully. Common hashing algorithms include MD5, SHA-1, SHA 256, and bcrypt. While MD5 and SHA-1 are considered outdated for password storage due to security weaknesses, modern algorithms such as bcrypt, scrypt, and

Argon2 provide much stronger protection because they are designed to resist password-cracking attacks.Hashing plays a vital role in protecting user credentials and reducing the risk of exposing passwords if a database is compromised.

5) Difference between Hashing and Encryption

Hashing and encryption are both security techniques used to protect information, but they work differently.

S.No	Hashing	Encryption
1	One-way process	Two-way process.
2	Cannot be reversed to original data.	Can be decrypted with a key.
3	Mainly used for password storage and file Integrity.	Used to protect confidentiality of data during storage and transmission.
4	Produces a fixed-length hash value regardless of input size.	Produces ciphertext whose size is generally depends on the input
5	Examples: MD5,SHA, bcrypt	Examples: AES, RSA, 3DES

6) Common Hash Types (MD5, SHA, bcrypt)

Different hashing algorithms are used in cybersecurity to protect passwords and verify data integrity. Some commonly used hash types are MD5, SHA, bcrypt.

MD5

- A legacy 128-bit hashing algorithm that is now cryptographically broken and vulnerable to collision attacks.
- It is obsolete for password storage but occasionally utilized for basic, non-security file integrity checks.

SHA (Secure-hash)

- A family of cryptographic hash functions designed to generate fixed-size outputs for data integrity verification.
- Older versions like SHA-1 are deprecated due to severe vulnerabilities, prompting a shift to stronger standards.

Bcrypt

- A specialized password-hashing function built on the Blowfish cipher that incorporates built-in salting mechanisms.
- It features a tunable work factor to intentionally slow down brute-force attacks, making it ideal for user credentials.

7) Common Encryption Algorithms (3DES, AES, RSA)

3DES

- A legacy symmetric-key block cipher that applies the original Data Encryption Standard (DES) cipher algorithm three times to each data block.
- It was created to enhance security over single DES, but is now considered deprecated and vulnerable to modern cryptanalytic attacks.

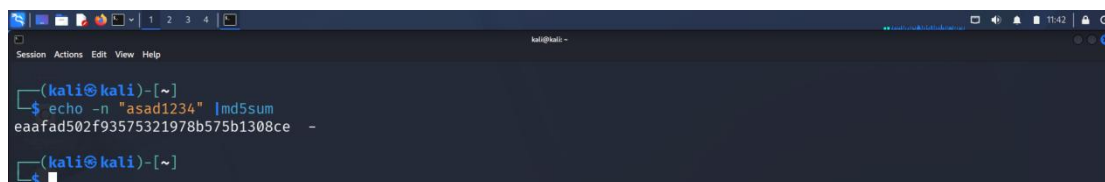
AES

- A widely adopted symmetric encryption standard that secures data using block ciphers with key lengths of 128, 192, or 256 bits.
- It is currently the global benchmark for secure communications, combining high-speed execution with robust cryptographic strength.

RSA

- A foundational asymmetric cryptographic algorithm that relies on the mathematical complexity of factoring large prime numbers.
- It uses a public-private key pair to securely handle digital signatures, key exchanges, and encrypted data transmission.

8) Practical Implementation of Creating an MD5 hash

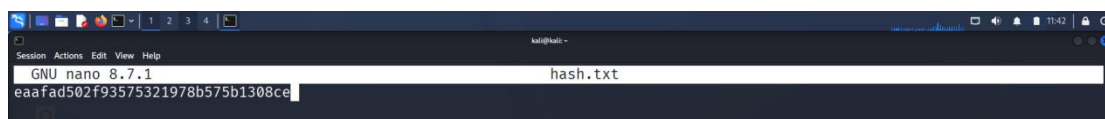


```
(kali㉿kali)-[~]  
$ echo -n "asad1234" |md5sum  
eaafad502f93575321978b575b1308ce -  
(kali㉿kali)-[~]  
$
```

Figure 1

In the **Figure 1** it is shown that **MD5 Hash** is generated for the Password **asad1234**.

Saving a Hash into File



```
GNU nano 8.7.1 hash.txt  
eaafad502f93575321978b575b1308ce
```

Figure 2

In the **Figure 2** it is shown that generated Hash is saved into File **hash.txt**.

Verifying the Hash

```
(kali@kali)-[~]
$ cat hash.txt
eaafad502f93575321978b575b1308ce
(kali@kali)-[~]
$
```

Figure 3

In the **Figure 3** hash is verified using cat command by reading the file **cat hash.txt**.

Cracking MD5 Hash using John the Ripper

```
(kali@kali)-[~]
$ john --format=Raw-MD5 --wordlist=/usr/share/wordlists/rockyou.txt hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (Raw-MD5 [MD5 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=2
Press 'q' or Ctrl-C to abort, almost any other key for status
asad1234 (?)
1g 0:00:00:01 DONE (2026-09-19 11:54) 0.5747g/s 5732Kp/s 5732Kc/s 5732KC/s asad4919546..asa14388
Use the "--show --format=Raw-MD5" options to display all of the cracked passwords reliably
Session completed.
(kali@kali)-[~]
$
```

Figure 4

In the **Figure 4** it is shown that the hash we generated for a password **asad1234** is successfully cracked using **john the ripper** tool with the help of dictionary **rockyou.txt** and received back the same password **asad1234**.

9) TryHackMe Room Crack the Hash

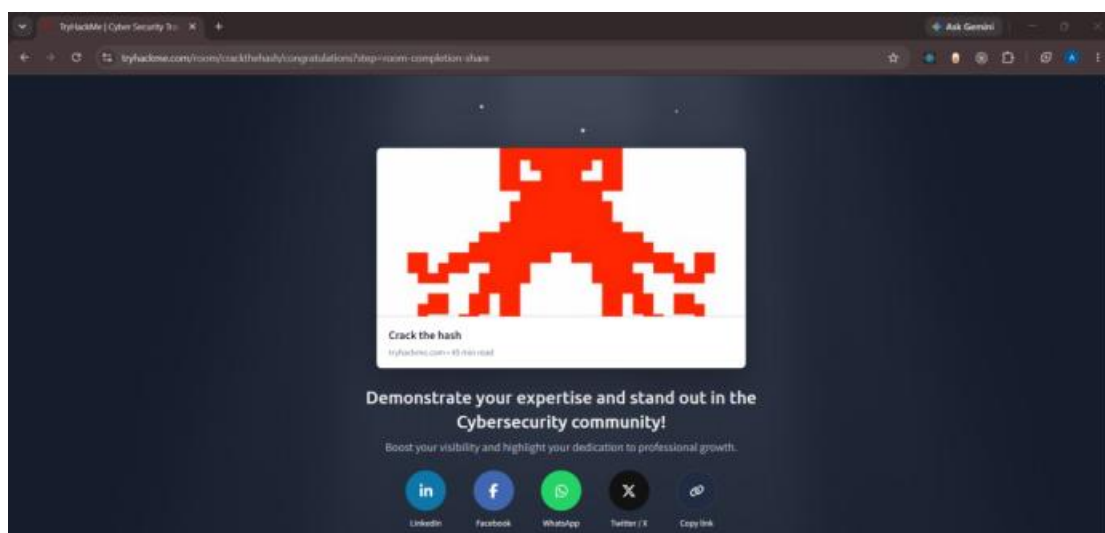


Figure 5

In the **Figure 5** it is shown that TryHackMe room **Crack the Hash** is completed Successfully. This TryHackMe room helped me understand different types of password hashes and how they can be tested in an authorized lab environment. I

learned how to identify hash types, use John the Ripper with a wordlist, verify cracked passwords, and understand the role of salts and stronger hashing algorithms. This practical improved my basic understanding of password security and hash cracking.

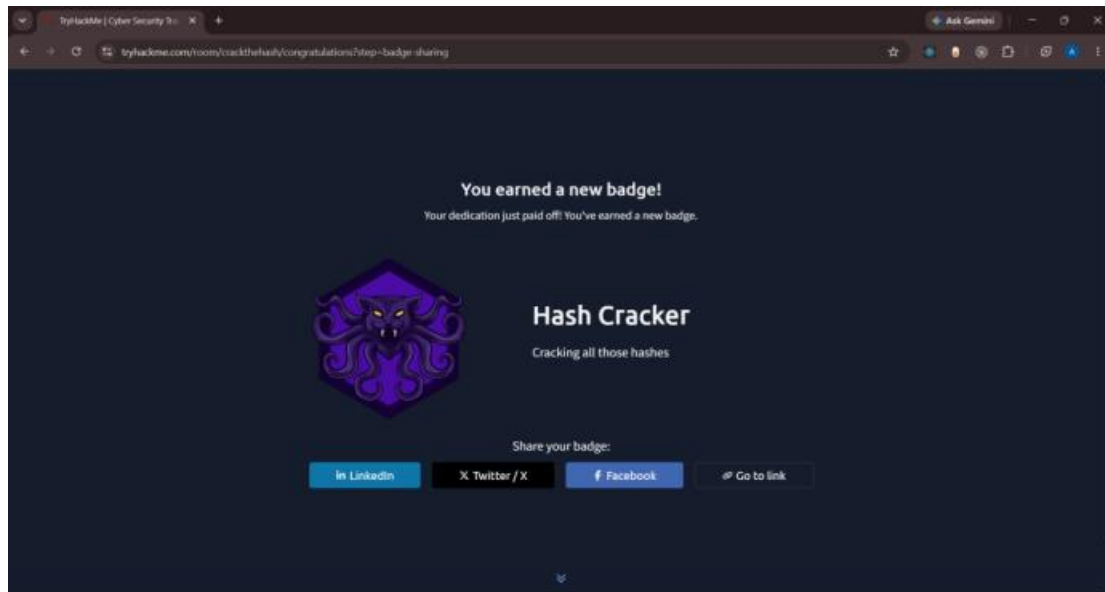


Figure 6

After completing the TryHackMe Crack the Hash room I have gained the badge Hash Cracker shown in **Figure 6** which reflects my understanding that how cracking hashes can reveal the password.

10) Conclusion

This practical helped me understand password hashing and password cracking in a practical way. I learned how to identify different hash types and use John the Ripper with the RockYou wordlist to crack hashes in the TryHackMe lab. I also learned about MD5, SHA-1, SHA-256, bcrypt, NTLM, and salted hashes. Overall, this task improved my basic understanding of password security and the importance of using strong passwords and secure hashing algorithms.

